
Computer Networks Lab

Week 02

Protocol Layer Simulation and wireless connections in Packet Tracer (Basic Ping & Traceroute)

Tools:

Cisco Packet Tracer 8.2+: Download from <https://www.netacad.com/resources/lab-downloads?courseLang=en-US>

Objectives:

- Master Packet Tracer Simulation Mode for protocol analysis and packet flow visualization
- Configure network hardware (routers, switches, hubs, PCs, wireless devices) using GUI and CLI
- Design and implement four network topologies (Star, Ring, Bus, Mesh) with protocol analysis
- Set up wireless networks with WPA2-PSK security protocol examination
- Create multi-network environments bridging wired and wireless segments
- Use diagnostic tools (ping, traceroute) for connectivity testing and path analysis
- Analyze protocol differences between Ethernet and wireless communications
- Troubleshoot network issues through systematic protocol-level investigation

Introduction to Protocol Layer Simulation

In computer networks, **protocols** are standardized rules that govern data transmission and reception, ensuring seamless communication between devices regardless of hardware or software differences. Common protocols include:

- **ARP (Address Resolution Protocol)** – Resolves IP addresses to MAC addresses.
- **ICMP (Internet Control Message Protocol)** – Sends messages and operational information.
- **HTTP (Hypertext Transfer Protocol)** – Facilitates web communication.
- **HTTPS (Hypertext Transfer Protocol Secure)** – Secures HTTP traffic with encryption.

Using Simulation Mode for Protocol Analysis

Packet Tracer's Simulation Mode allows you to:

- **Visualize packet flow** through the network step-by-step
- **Examine protocol headers** at each layer
- **Observe encapsulation/decapsulation** process
- **Analyze timing** and packet interactions
- **Filter specific protocols** for focused analysis

To enable Simulation Mode:

1. Click the **Simulation** tab at bottom-right of Packet Tracer
2. Use **Event List Filters** to select which protocols to display
3. **Play Controls** let you step through packet transmission

ICMP and Basic Ping Analysis

The **ping command** uses **ICMP (Internet Control Message Protocol)** to test network connectivity.

When you ping a destination:

1. **ICMP Echo Request** is sent to the destination
2. Destination responds with **ICMP Echo Reply**
3. **Round-Trip Time (RTT)** is measured and displayed
4. Statistics show **packet loss** and **delay variations**.

ICMP operates at the **Network Layer (Layer 3)** and provides:

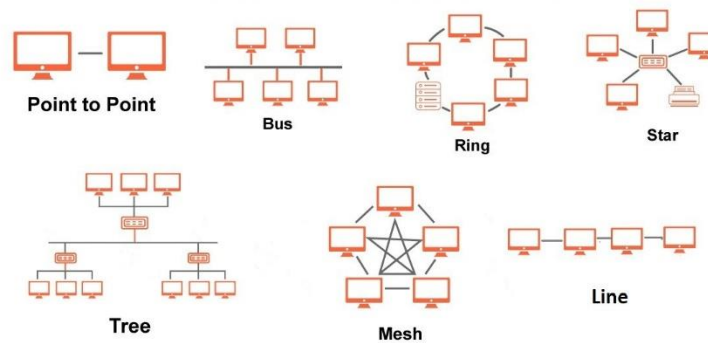
- **Connectivity testing** through echo requests/replies
- **Error reporting** when packets cannot be delivered
- **Network troubleshooting** information

Introduction to Network Topologies

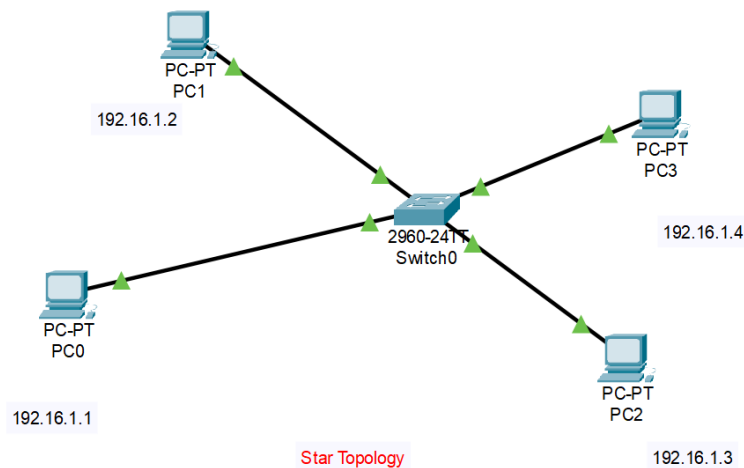
A network topology defines how devices (nodes) and links are arranged in a computer network, affecting data flow, reliability, and scalability.

This lab focuses on four common topologies: star, ring, bus, and mesh. Each has unique characteristics that impact network performance and troubleshooting.

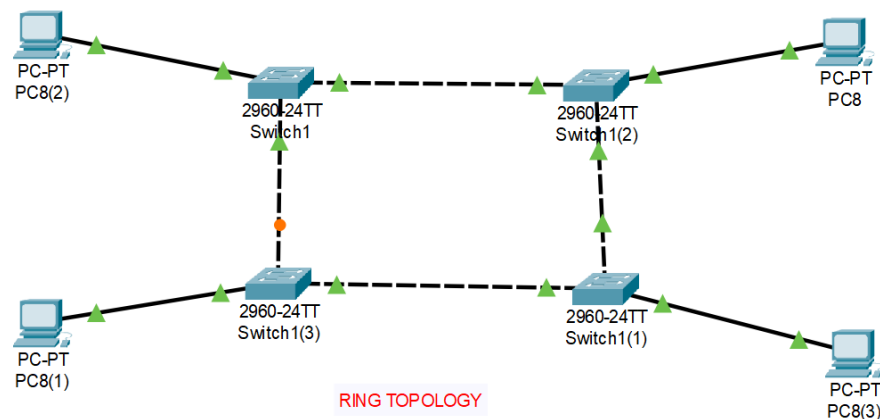
Network Topology Types



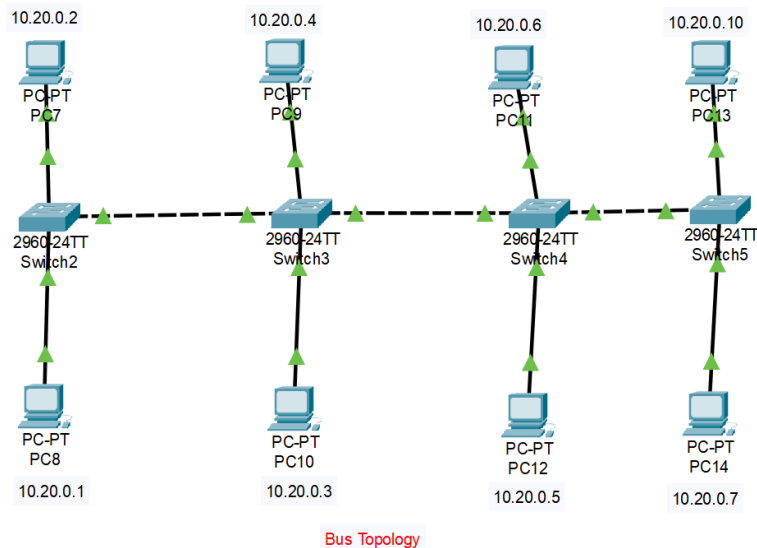
- **Star Topology:** Devices connect to a central switch or hub. Data passes through the central device, making it easy to manage but dependent on the central device's reliability. Common in home and office networks.



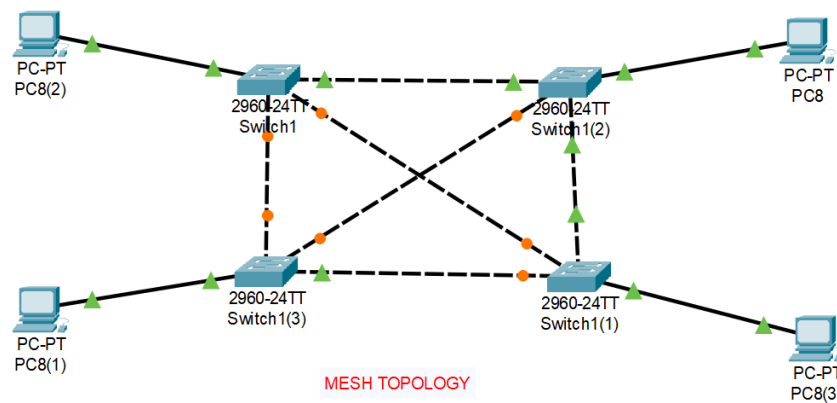
- **Ring Topology:** Devices form a circular network, with each device connected to two others. Data travels in one direction, reducing collisions but risking network failure if one link breaks.



- **Bus Topology:** Devices connect to a single shared cable (bus). Data broadcasts to all devices, making it simple but prone to congestion and single-point failures.



- **Mesh Topology:** Devices are interconnected, allowing multiple data paths. This increases reliability but requires more cabling and configuration.



In this lab, you'll simulate these topologies in Cisco Packet Tracer and analyze their behavior using Simulation Mode.

Setting Up a Wireless Connection

This section guides you through configuring a wireless network in Cisco Packet Tracer using a **WRT300N wireless router**, which is ideal for basic wireless configurations and supports the required settings for this lab. and **two PCs** with wireless modules.



1. Add Devices

- From the Wireless Devices section, drag a WRT300N wireless router to the workspace.
- From the End Devices section, add two PC-PT devices.

2. Configure the Router

- Click the router, go to the GUI tab, and disable the DHCP server to use static IP addresses.
- In the Wireless tab, set the SSID to *labtwo*.
- In the Wireless Security tab, select **WEP**, set the password to **0123456789**, and save.

3. Configure PCs for Wireless

- For each PC, go to **the Physical tab**, power off the device, remove the Ethernet module, and add a **WMP300N wireless module**. Power the PC back on.
- In the **Desktop tab**, open **PC Wireless**, scan for networks, select *labtwo*, and enter the password **0123456789**. Verify the connection status.

4. Assign IP Addresses

- On **PC1**, go to **Desktop > IP Configuration**, select Static, and set: **IP: 192.168.0.5, Subnet Mask: 255.255.255.0, Gateway: 192.168.0.1**
- On **PC2**, set **IP to 192.168.0.6**, keeping the same subnet mask and gateway.
- On the router, ensure the wireless interface is set to **192.168.0.1/24**.

5. Test Connectivity

- On **PC1**, open Command Prompt and run **ping 192.168.0.6**.
- Verify replies from **PC2**, indicating a successful wireless connection.
- Take a screenshot of the ping output and the network topology with devices labeled.

By following these steps, you will have successfully created and tested a wireless connection in Cisco Packet Tracer, allowing the two PCs to communicate wirelessly through the WRT300N router.

Traceroute for Path Analysis

The *traceroute* command on a Cisco device can be used to identify the path used by a packet to reach its target. It identifies all the routers in the path from the source host to destination host and it can be useful when troubleshooting network problems. Using this command you can figure out which router in the path to an unreachable target should be examined more closely as the probable cause of the network's failure.

It works by sending a series of **ICMP** Echo Request packets (also known as **ping** packets) with incrementally increasing TTL values (Time To Live). The TTL is a field in the IP header that limits the number of hops (routers) a packet can traverse before being discarded. Here's how it operates:

- Sending packets with increasing **TTL (Time To Live)** values
- Each router decrements TTL and sends **ICMP Time Exceeded** when TTL=0
- This reveals **each hop** in the path and **delay to each router**

Commands:

- **Windows:** `tracert <destination>`
- **Linux/Unix:** `traceroute <destination>`

How to Access Different Interfaces:

In Packet Tracer PCs (Windows-like):

1. Click on PC → Desktop tab → Command Prompt
2. Type: `tracert <destination_ip>`
3. Press Enter and observe the output

In Real Cisco Routers/Switches:

1. Access CLI through console or SSH
2. Type: `traceroute <destination_ip>`
3. View hop-by-hop path information

Key Differences:

- **Windows:** Uses `tracert` (short form)
- **Cisco/Linux:** Uses `traceroute` (full form)
- **Packet Tracer PCs:** Simulate Windows environment, so use `tracert`

Lab Tasks

Task 1: Identify and Label Network Hardware with Protocol Analysis (10 Marks)

Objective: Set up a network in Cisco Packet Tracer with a router, switch, hub, PCs, laptop, and wireless router, label all devices, and analyze packet behavior through a switch versus a hub using Simulation Mode.

1. Add Devices

- One 1841 Router (central routing device for interconnecting networks).
- One 2960-24TT Switch (connects devices within the same network).
- One Generic Hub (demonstrates a shared LAN connection).
- Two PC-PT devices (end-user computers for wired connectivity).
- One Laptop-PT (portable device for wireless connectivity).
- One WRT300N Wireless Router (source of the wireless network).

2. Connect Devices

- Connect PC1 to the Switch (e.g., FastEthernet0/1) using a straight-through cable.
- Connect PC2 to the Hub (e.g., Ethernet0) using a straight-through cable.
- Connect the Laptop to the Wireless Router wirelessly (configure in part 4).
- Connect the Switch to the Router (Router's FastEthernet0/0 to Switch's FastEthernet0/2) using a straight-through cable.
- Connect the Hub to the Router (Router's FastEthernet0/1 to Hub's Ethernet1) using a straight-through cable.
- Connect the Wireless Router (LAN port) to the Switch (FastEthernet0/3) using a straight-through cable.

3. Configure IP Addresses

- **PC1** IP Address: **192.168.1.10**, Subnet Mask: **255.255.255.0**, Gateway: **192.168.1.1**
- Configure **PC2** with same steps.
- Configure **Laptop** with same steps after wireless configuration.
- Enter Router's global configuration mode and configure the interfaces in **CLI mode**. Configure for both interfaces here is the example for one:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
```

4. Configure Wireless Connectivity

- Go to GUI tab, disable DHCP.
- Under Setup, set IP address to 192.168.1.3/24.
In **Wireless tab**, set SSID to lab2_wifi, security to **WPA2**, password to **secure123**.
- Connect Laptop with the Wireless Router.

5. Label Devices

- Label all the devices and their IP addresses along with them.
- Use the annotation tool to add your roll number, name and section to the workspace.

Submission:

- Enter Simulation Mode, and perform ping tests from different end devices, and analyze packet behavior. There will be **errors in this configuration** you have to IDENTIFY that and write down your findings.
- Write 2–3 sentences explaining how the switch's unicast forwarding differs from the hub's broadcast behavior.
- A screenshot of the complete topology with all devices labeled (name, function, roll number) should be attached in the PDF file.

Task 2: Create Network Topologies with Protocol Layer Simulation Analysis (5 Marks)

Objective: Simulate and analyze star, ring, and bus topologies using Cisco Packet Tracer, focusing on protocol behavior in Simulation Mode.

General Setup:

- Use four PC-PT devices and appropriate network devices for each topology.
- Label all devices with their name and function (e.g “PC1: End-user device”).
- Take screenshots of each topology and Simulation Mode outputs, ensuring your roll number is visible (use Packet Tracer's annotation tool).

1. Star Topology

Add a **2960-24TT** Switch and connect four PCs to it using Ethernet cables.

Assign IP addresses: **PC1:** 192.168.1.1/24, **PC2:** 192.168.1.2/24, **PC3:** 192.168.1.3/24, **PC4:** 192.168.1.4/24.

Protocol Analysis Steps:

- **Enable Simulation Mode**, filter for **ICMP** and **ARP** protocols.
- From PC1: ping 192.168.1.3 -n 1

- Step through simulation and examine:
 - **ARP Request/Reply** packets (if needed for MAC address resolution)
 - **ICMP Echo Request** packet details (Type, Code, Sequence number)
 - **ICMP Echo Reply** packet details
 - **Frame encapsulation** at each OSI layer
- **Document:** How the switch handles unicast traffic vs broadcast traffic.

2. Ring Topology:

- Add four **2950-24 switches** and connect them in a loop (Switch1 to Switch2, Switch2 to Switch3, etc.).
- Connect one PC to each switch (PC1 to Switch1, etc.).
- Assign IP addresses: **PC1:** 192.168.2.1/24, **PC2:** 192.168.2.2/24, **PC3:** 192.168.2.3/24, **PC4:** 192.168.2.4/24.

Use ping command to test connectivity and observe data flow patterns.

3. Bus Topology:

Create a bus topology with 4 devices (using switches to simulate). Assign IP addresses:

- PC1: 192.168.3.1
- PC2: 192.168.3.2
- PC3: 192.168.3.3
- PC4: 192.168.3.4

Test connectivity using ping command.

4. Mesh Topology with Traceroute Analysis:

Create a mesh topology where each PC is connected to every other device. Assign IP addresses:

- PC1: 192.168.4.1
- PC2: 192.168.4.2
- PC3: 192.168.4.3
- PC4: 192.168.4.4

Enhanced Analysis:

- Use ping command to test basic connectivity
- Use tracert command from PC1 to PC4 to observe the direct path
- **Compare** with a simulated "longer path" scenario
- **Document:** How mesh topology provides redundant paths

For each topology, document one advantage or disadvantage based on your **protocol-level observations** from simulation mode.

Task 3: Wireless Network with Multi-Network Traceroute Analysis (5 Marks)

Objective: Configure a wireless network and analyze protocol behavior across different network segments using traceroute.

Network Setup:

1. Create Multi-Network Topology:

- **Network 1 (Wired):** One PC connected to a 2960 switch
- **Network 2 (Wireless):** WRT300N wireless router with two wireless laptops
- Connect the switch to the WRT300N's LAN port to bridge networks

2. IP Configuration:

- **Wired Network (192.168.1.0/24):**
PC1: 192.168.1.10/24, Gateway: 192.168.0.1
- **Wireless Network (192.168.0.0/24):**
 - Laptop1: 192.168.0.10/24, Gateway: 192.168.0.1
 - Laptop2: 192.168.0.11/24, Gateway: 192.168.0.1
 - WRT300N: 192.168.0.1 (wireless) and 192.168.1.1 (LAN)

3. Wireless Configuration:

- **SSID:** "cnet_lab2"
- **Security:** WPA2-PSK
- **Password:** "lab2secure"

Protocol Analysis Requirements:

A. Wireless Protocol Simulation:

- **Enable Simulation Mode** with ICMP, and ARP filters
- From Laptop1: ping 192.168.0.11 -n 1 (wireless-to-wireless)
- **Examine wireless frames** in simulation mode.

B. Cross-Network Traceroute Analysis:

- From PC1 (wired): tracert 192.168.0.10 (to wireless laptop)
- From Laptop1 (wireless): tracert 192.168.1.10 (to wired PC)
- **Record hop information** showing the router as intermediate device

Documentation Required:

- Network topology screenshot showing both wired and wireless segments
- **Simulation mode screenshots** of wireless frame details
- **Traceroute outputs** showing multi-hop paths
- **Protocol analysis** comparing wired vs wireless communication
- Brief description of configuration process and **observed protocol differences**

Submission Guidelines:

- Submit a single PDF document containing all tasks and the CISCO packet files to the GCR, **ensure your file name is visible in all pictures and contains your roll number**.
- Ensure that all screenshots are clear and labeled appropriately.
- Include your name, roll number, and lab number inside the **first page** of the document.
- Name the submitted document as follows: SectionLetter_RollNumber_LabNumber, for example: **J_i221234_Lab1**.
- Name the CISCO files as follows: SectionLetter_RollNumber_TaskNumber, for example: **J_i221234_T1**.
- In the demo, there will be deductions based on your answers to the questions asked.

NOTE:

- You must show clear screenshots of every step you take. Marks will be deducted for this mistake and in some cases you may be awarded 0 (this applies to all tasks).
- Screenshots where your name, roll number and section are NOT visible, will automatically be given 0.

Sample screenshot:

